AMOXICILLIN + CLAVULANIC ACID

Penicillin allergy

Take at start of meal

CHEMICAL NAME Amoxicillin (aminopenicillin) + potassium clavulanate: (2R,3Z,5R)-3-(2-hydroxyethylidene)-7-oxo-4-oxa-1-azabicyclo[3.2.0]heptane-2-carboxylate

CLASS Aminopenicillin + β-lactamase inhibitor (co-amoxiclav)

TRADE NAMES Augmentin · Curam · Clavulin · Megamox

THERAPEUTIC USES / INDICATIONS

- Infections caused by β-lactamase-producing organisms: sinusitis, otitis media, lower respiratory tract infections.
- Animal and human bites; skin and soft-tissue infections; diabetic foot (mild).
- Complicated UTI, dental abscess.
- Intra-abdominal infection and surgical prophylaxis (IV).

ADMINISTRATION

- **PO:** at the start of a meal (improves clavulanate absorption, reduces GI upset).
- **IV:** slow injection over 3–4 min or infusion over 30–40 min.
- Do not substitute tablet strengths (clavulanate content differs).

DOSAGES

- **Adult PO:** 625 mg (500/125) every 8 h, or 1 g (875/125) every 12 h.
- **Adult IV:** 1.2 g every 8 h.
- **Children PO:** 20–45 mg/kg/day (amoxicillin component) divided every 8–12 h; high dose 90 mg/kg/day.
- Max clavulanate 375 mg/day in adults (PO); CrCl <30: avoid 875/125.

PHARMACOKINETICS

- **Peak:** 1–1.5 h.
- Amoxicillin renally excreted unchanged; clavulanate metabolised in liver.
- **t½:** ~1 h each.

PHARMACODYNAMICS

- Amoxicillin inhibits **cell-wall synthesis** (bactericidal).
- Clavulanic acid irreversibly inhibits β-lactamases → extends spectrum to *H. influenzae*, *M. catarrhalis*, *Staph. aureus* (MSSA), anaerobes.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea (more than amoxicillin alone), nausea, vomiting.
- Rash, candidiasis.
- Indigestion.

ADVERSE EFFECTS (SERIOUS)

- Cholestatic jaundice / hepatitis (clavulanate); more in elderly men, may appear up to 6 weeks after stopping.
- Anaphylaxis.
- *C. difficile* colitis.
- SJS (rare).

CONTRAINDICATIONS & CAUTIONS

- Penicillin allergy.
- History of co-amoxiclav- or penicillin-associated jaundice.
- **Caution:** hepatic impairment, renal impairment.

KEY INTERACTIONS

- **Warfarin:** ↑ INR.
- **Methotrexate:** ↑ toxicity.
- **Allopurinol:** ↑ rash risk.
- **Mycophenolate:** ↓ levels.

NURSING MANAGEMENT

- Check allergy history; observe after first dose.
- Give PO at the start of meals.
- Monitor LFTs in long courses or elderly; watch for jaundice.
- Monitor bowel pattern.
- Review culture results for de-escalation.

PATIENT EDUCATION

- Take at the beginning of a meal.
- Complete the course.
- Report yellow eyes or skin, dark urine (even after finishing).
- Report severe diarrhoea.

KEY POINTS

- Clavulanate inhibits β-lactamase.
- Cholestatic jaundice risk.
- Broader spectrum incl. anaerobes, MSSA.
- Adult IV 1.2 g every 8 h.

SOURCES CHECKED **[106]** Co-amoxiclav 125/31.25 mg/5 ml Powder for Oral Suspension (UK SmPC)
[107] Co-Amoxiclav 1000 mg/200 mg Powder for Solution for Injection/Infusion (UK SmPC) **[108]** Amoxicillin and Clavulanate Potassium, oral (US label)
[109] Amoxicillin and Clavulanate Potassium, oral (US label, REMEDYREPACK INC)

Full references p. 432–457